

Paul Allen School Computer Science Engineering, University Washington

University of California, Berkeley Sensitivity remains

We also show that for flat performance, the number of test-shot examples would benefit from repeating performance

prompting, parse instructions, and use a prompt format and a pre-defined prompt template. We propose a novel framework, **PREP**, that systematically analyzes and synthesizes prompts across plausible prompt formats, instead of the currently standard practice of finding a single prompt format. PREP rapidly evaluates a large number of prompt formats for a given task and format, reports the interval of expected performance without accessing model weights, and provides a detailed analysis of the prompt format and the model's behavior.

As large language models (LLMs) are adopted, fundamental component language technologies, crucial accuracy points when evaluated using LLaMA-2-13B

INTRODUCTION

As the capabilities of LLMs have rapidly improved, their sensitivity to input prompt features has been used to optimize performance via prompt engineering (White et al., 2023).

Furthermore, we present suite analyses that characterize the nature this sensitivity, including exploring the influence particular atomic perturbations and the internal representation particular formats.

design process includes making choices about wording, choosing few-shot examples for in-context learning, and making decisions about seemingly trivial features like formatting prompt, interpretation ways, with accuracy differences up to 76 points for LLaMA 2-13B. We also show that this variance is not eliminated by adding few-shot examples, increasing model size, instruction tuning.

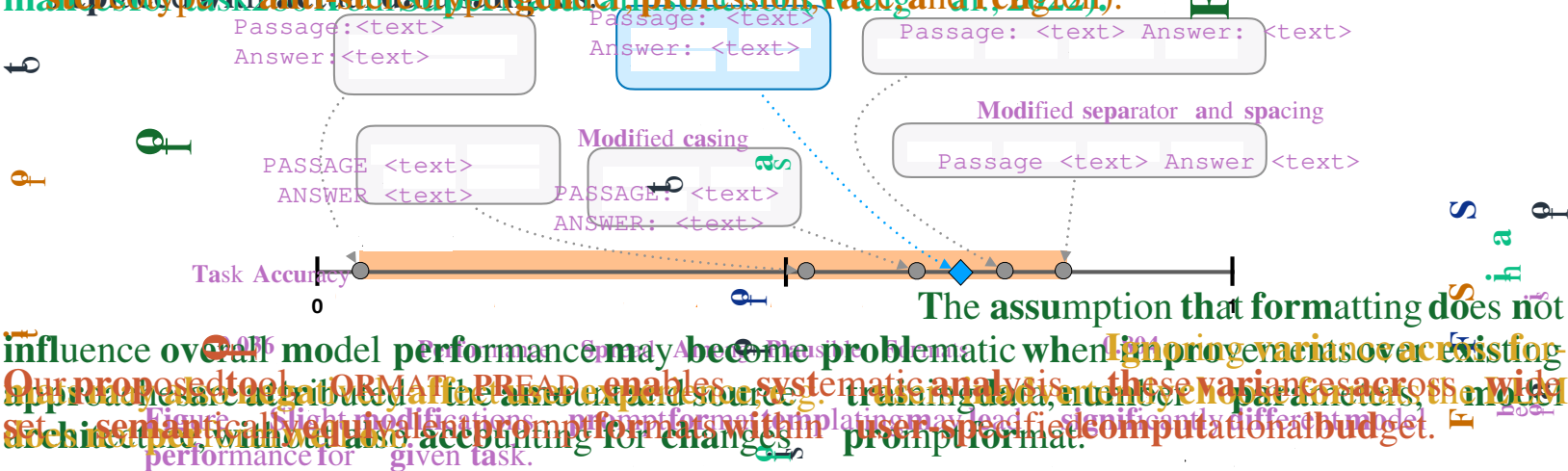
However, some anecdotal evidence points to formatting choices actually having a significant influence on model behavior (Aghajanyan, 2023).

often even the resulting templates, rarely reported discussed research papers, under the as-
sumption that performance variance across these choices insignificant compared variance across
data points models.

This process, and

iv:23

(Nadeem et al., 2021) require that each class has a different classifiable hit, and the task defines a mapping from the hit to the class. For example, the task of identifying the gender of a person from a text passage requires that the hit is the person's name, and the class is the gender.



The sensitivity of formatting choices that we discover across widely-used, open-source models suggests that future research would benefit from reporting performance spread over sufficient sample sizes, instead of assuming a single performance value. **ORMAT-PREAD** efficiently explores the space of prompt formats under user-specified computational budget using Bayesian optimization. We show the influence of formatting choices only weakly correlates between models, thus making fully exploring the space of prompt formats intractable. Computation costs scale linearly with the number of formats considered.

We find that choices of formatting few-shot examples during in-context learning introduces spurious biases that may lead to significantly different conclusions about model performance.

Among other results, we show that the separability of continuous prompt embeddings correlates with the spread observed in task performance. **ORMAT-PREAD** uses Bayesian optimization to identify the expected performance range with less than a computational cost of \$4.5, all without requiring access to model weights, which enables use on API-gated models and validation spread up to 56 accuracy points with GPT-3.5 and 64 accuracy points with GPT-4 across 320 formats and 53 tasks at a cost under 10USD on average per task.

We evaluate LLM performance over the space of prompt formats that may plausibly be chosen by a non-adversarial user when designing a prompt for a target task, where the space is defined by a grammar (§3.1). **ORMAT-PREAD** uses Bayesian optimization to identify the expected performance range with less than a computational cost of \$4.5, all without requiring access to model weights, which enables use on API-gated models and validation spread up to 56 accuracy points with GPT-3.5 and 64 accuracy points with GPT-4 across 320 formats and 53 tasks at a cost under 10USD on average per task.

We evaluate LLM performance over the space of prompt formats that may plausibly be chosen by a non-adversarial user when designing a prompt for a target task, where the space is defined by a grammar (§3.1).



Published conference paper ICLR 2023. We find UCB is frequently applied in the literature, assigned the credit of being effective for prompt optimization.

By default, each model is evaluated with 100 prompts sampled from the upper confidence bound.

CHARACTERIZING PROMPT FORMAT VARIANCE WITH PROMPT READ Naive Sampling.

EXPERIMENTAL SETUP

Data.

We use Super-NaturalInstructions (Wang et al., 2022) with diverse human-written formats and instructions comprising multiple choice task and classification tasks with basic fields.

Datasets are assumed size 1,000 for fair evaluation across tasks.

Few-shot examples are chosen randomly from each dataset and used for evaluation.

We evaluate LLaMA-2-7B (Ba et al., 2023), GPT-3.5-Turbo (OpenAI, 2022), and autoregressive LMs.

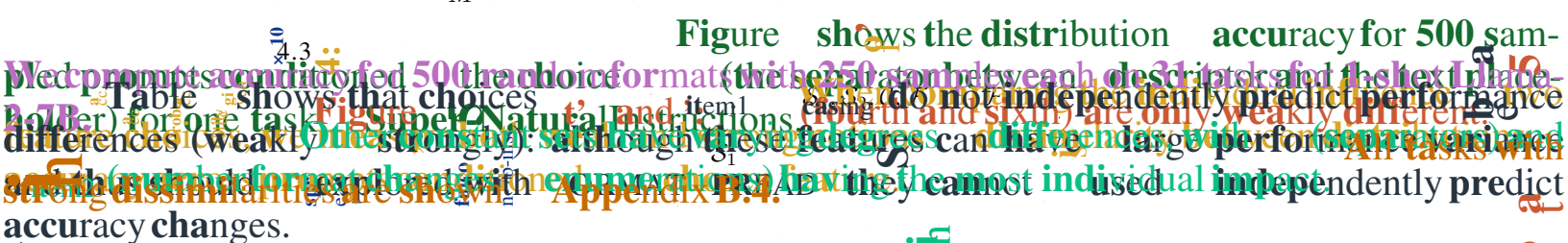
We use two popular measures for computing accuracy: exact prefix matching and probability ranking.

Models.

Appendix B.2 shows additional analysis. Rankixagaprefixmatching, which predicts even higher than those in Section 4.2. Results indicate that performing rankixagaprefixmatching affects results significantly (the models’ output distribution).

PROMPT FORMATS HAVE LARGE PERFORMANCE SPREAD NOT ELIMINATED BY INSTRUCTION TUNING. We observe several tasks with performance spread over accuracy points. Because this analysis uses only randomly sampled formats, it represents lower bound of the true spreads for all tasks. The mode about the number of few-shot examples, and least points for all Falcon settings. For each evaluation task we randomly sample plausible prompt formats and use ORMAT PREAM compute performance spread for each modeling and -shot choice (Figure 9).

Furthermore, there exists significant performance spread regardless increased model size (Figure 10 and Figure 11 for Llama-2-70B), instruction tuning (Figure 12), or non-classification tasks (Figure 13). Comparison trends between models are often reversed just by choosing different formats. The entire accuracy distributions samples with % displayed respectively for the two model comparisons are often reversed: LLaMA-2-13B and -70B reverse trend by least trials. We use one-sided McNemar tests, also known as paired z-tests, since we evaluate models on the same set of samples. We test the significance accuracy being better than differential and being worse than under



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Table Tasks. We use at least two constants yield different performance, and weakly different their boxes boxplot not overlap, strongly Strong Diff. boxes including whiskers not overlap).

task280 passage: answer: passage
task317 Passage:: Answer::
task190 Sentence[I]- Sentence[I]-
task904 Answer
task320 input:: output::
task322 target
task279 COMMENT: ANSWER: Separator Used
Passage

Figure Example accuracy variance for different choices constants for task1283.

Task Prompt Format Prompt Format

0.55
0.50
0.45
0.40

0.043 0.826 0.783
0.076 0.638 0.562
0.360 0.614 0.254

0.144 29% 1%
0.132 43% 0.616 22% 0.198
0.238 0.361 0.476 0.115
0.176 0.614 0.714 0.100
0.173 45%
0.188 3% 0%

yields higher accuracy than 4 for tasks.

Table Examples atomic changes' impact accuracy using probability ranking (prefix matching shown Table 4).

Table 4

p_1 p_2 p_1 p_2 a

$\emptyset \setminus \emptyset$ $\emptyset$ $\emptyset \setminus \emptyset$ $\emptyset$ a

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multiple-choice and non-multiple-choice tasks respectively.

non-multiple choice), sample 300
exact prefix matching) each

$(p_i \text{ samples})$
 p_i 250 32.4

triples for each, and the compute accuracy (using

triple will result monotonicity 33.3% the time, this suggests that local search mechanisms like simulated annealing may not effective they require locally smooth search space.

For each task, and for both -shot settings, we collect prompt embeddings from LLaMA-2-7B. We demonstrate that format choice yields highly identifiable transformation over this embedding, corresponding that randomly sampled valid formats for the evaluation examples.

transformations on the output probability distribution.

$\{1, 5\}$

10

1000

We represent prompt embeddings the last hidden layer obtained when processing the whole input prompt (immediately before generating the first token).

PROMPT FORMATS ARE IDENTIFIABLE TRANSFORMATIONS OF PROMPT EMBEDDINGS

31

Prompt format choices represent deterministic transformations of the input even if the resulting performance is low. We find that although the original prompt embeddings are not principal components, the input even if the resulting performance is low.

AST EXPLANATION OF THE PROMPT FORMATTING SPACE

FORMAT PREAD

4.5

Section 4.2, we demonstrate that even when sampling just 10 formats from the space plausible formats, we still observe significant performances spread on many tasks.

2.1%

3
4
5

7

We find that classifier accuracy given just the top two components correlates moderately with the spread performance the prompts they represent

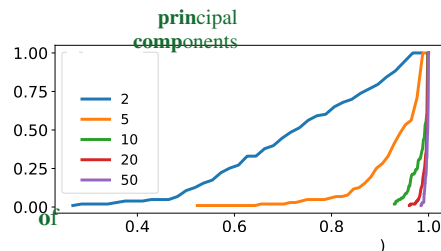
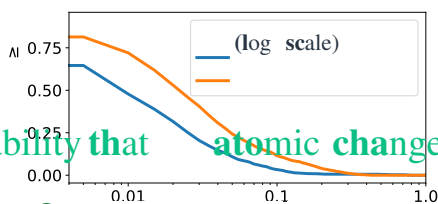
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Figure 4.2: We find that classifier accuracy given just the top two components correlates moderately with the spread performance the prompts they represent

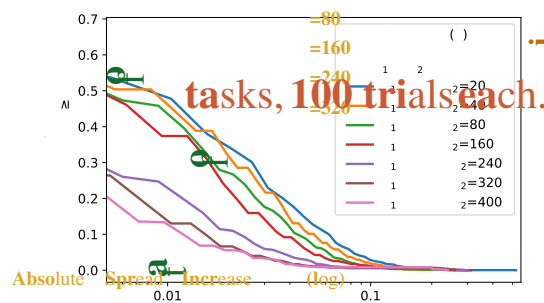
Figure 4.2: We find that classifier accuracy given just the top two components correlates moderately with the spread performance the prompts they represent

tasks, this sampled atomic changes each, given impact accuracy for two scoring criteria.

Figure (e.g. Probability that atomic change



Absolute Accuracy Diff. Spread Increase when increasing sampling from formats





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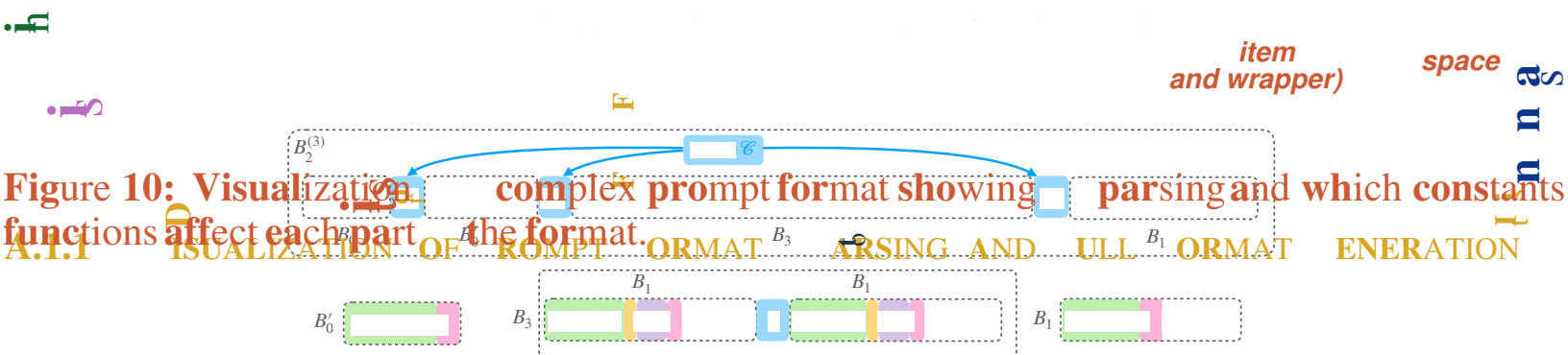
GRAMMAR DEFINITION AND INSTANTIATION DETAILS

A.1 EQUIVALENCE RELATION possible that generated formats equivalent their string representation and are not equivalent according to this equivalence relation.

$$B_0(d_i, s_i) \quad d_1 = d_2; \quad p_i = B_1(d_i, s_i, j_i) \quad d_1 = d_2; \quad p_i = B_2^{(n)}(X_{1,i}, \dots, X_{n,i}, c_i)$$

Precisely, $X_{j,1} \sim X_{j,2} \forall 1 \leq j \leq n; \quad p_i = B_3^{(n)}(d_i, j_{1,i}, \dots, j_{n,i}, s_1, s_2, c, f) \quad d_1 = d_2$ and only least one of the following hold:

Figure 10 shows visualization how complex format parsed using our defined grammar.



instruction was Given sentence and two words that appear in it, answer which one of the two (A or B) appeared first in the sentence.

Full prompt consists of instruction, few-shots and data point solve.

Over the lazy dog

OPTIONS: CHOICE (A) <text> CHOICE (B) <text> ANSWER: <text>

CHOICE (A): lazy CHOICE (B): dog

ANSWER: Given sentence and two words that appear in it, answer which one of the two (A or B) appeared first in the sentence.

FORMAT PREAD forces all instantiations multiple choice variable change jointly maintain coherence, and this includes text the instruction.

Note that we always use space character between instruction and few-shots.

where the Options do not have corresponding field.



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on

Bright alter while ex Gouts d value

This restricts the parameter

$$\beta = 5.$$
$$\alpha > 1 \quad \beta > 1.$$
$$\alpha >$$

$$= \alpha \cdot x + \beta \cdot x$$
$$\frac{\beta \cdot x}{1 - x}$$

(α, β)

Since we want to avoid the model being overconfident, we use a heuristic to filter out tasks whose instructions are too long (over 3,000 characters) and datasets with invalid outputs. We also filter out tasks whose instructions are too long (over 3,000 characters) and datasets with invalid outputs.

ADDITIONAL EXPERIMENTS AND INFORMATION AND LOTS

B.1 TASK SELECTION

We use number heuristics filter Super-NaturalInstruction tasks our set evaluation

NaturalInstruction tasks where we included format one was not provided the dataset. We also removed tasks where we found prior We did not include more than 0% accuracy from the same original dataset. Some Super-NaturalInstruction tasks are derived from the same original dataset, but ask different questions.



The selected tasks are: task114, task190, task296, task323, task385, task1186, task1419, task065, task161, task069, task070, task163, task286, task322, task327, task328, task335, task337, task580, task607, task608, task609, task904, task905, task1283, task1284, task1297, task1347, task1387, task1420, task1421, task1423, task1502, task1612. Llama-2-70B was compared to Llama-2-7B on these tasks. Llama-2-70B spreads.

{Llama-2-7b, Llama-2-13b}
{Llama-2-7b, falcon-7b}
{falcon-7b, falcon-7b-inst}

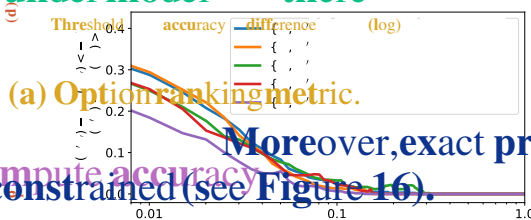
B.2 ADDITIONAL RESULTS FOR ACTION

Llama-2-7b	Llama-2-13b	M^P	57.46%
Llama-2-7b	Falcon-2-7b	M^P	55.91%
Falcon-7b	Falcon-7b-Inst	M^P	61.11%

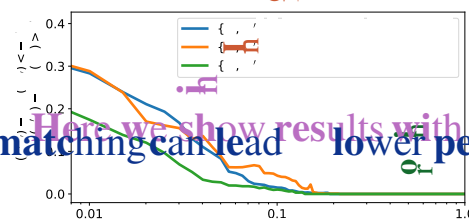
Figure 12: Probability prompt being worse than least points using model given that prompt was better than prompt on using model

Table shows that format has lower performance than format under model there

(b) Exact prefix matching metric.



(a) Option ranking metric.



Formats are not inherently good than format under model there

Moreover, exact prefix matching can lead lower performance generation less constrained (see Figure 16).

Often, failures prefix matching are associated with degeneration, i.e., cases where the model does not answer any the valid options, motivating the use ranking experiments with exact prefix matching accuracy. Degeneration makes models (especially smaller models) more unlikely have high accuracy out the box.

that findings format being better This finding further supported Figure 12, which shows worse than another are often inconsistent across models.

This weak relative order preservation suggests that prompt format performance model may not extrapolated different model, other words, that there are inherently good bad format. Sometimes when model does not generate the correct answer with exact prefix matching, also does not generate valid response, i.e., degenerates.

results shown for probability ranking. Section 4.2, Figure 15 shows that the probability reversing performance trend between two models just by changing prompt remains high when using exact prefix matching. This further motivates the need for running ORM re-PRPAD when benchmarking models with this accuracy metric. This increased spread may partly due degeneration, we will detail next.

Table shows examples atomic changes yielding large accuracy changes with exact prefix matching metric.

This suggests that very often when model does not return valid answer, does not return any valid answer

Table shows that the correlation between accuracy and centered mass moderate high depending on the model. Given model, task, and format, let the centered mass the ratio we will now quantify this output phenomenon using 53 SuperOptima (logit-normal classification and multiple choice tasks).

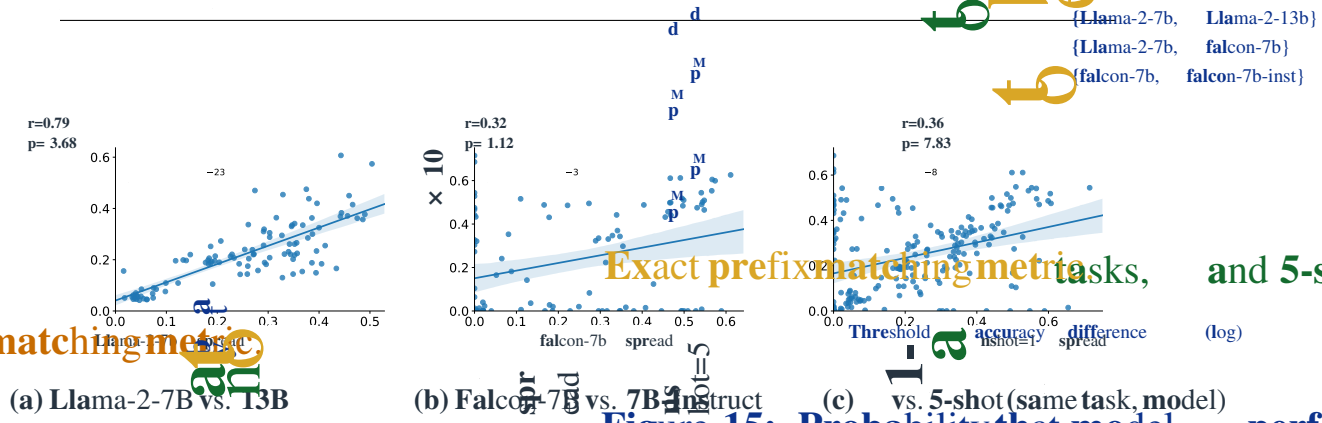
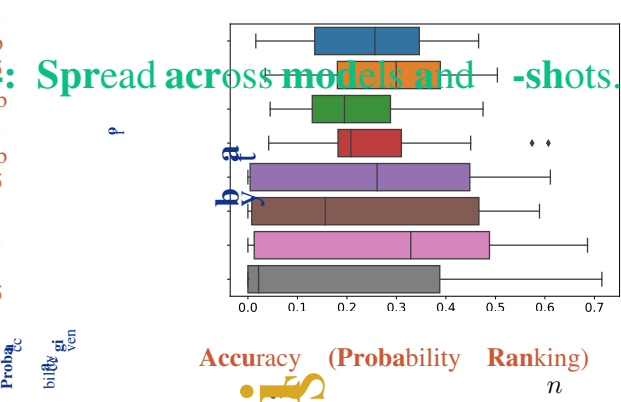
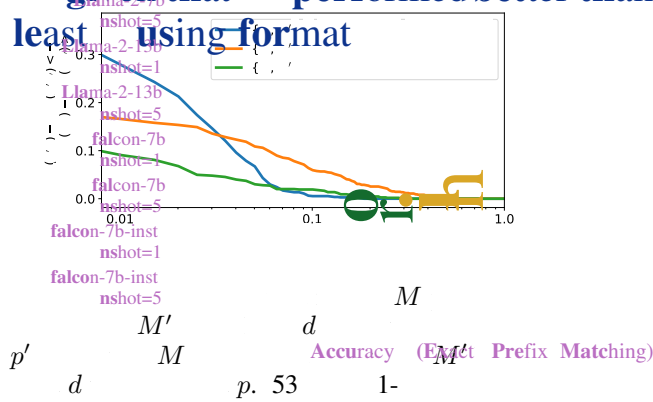


Figure 13: Spread comparison between evaluating the same task under different models. The plots show the relationship between accuracy and spread for different model pairs and shot counts.



(a) Accuracy boxplot for the selected Natural-Instruction tasks, option ranking metric.



(b) Accuracy boxplot selected Super Natural-Instruction tasks, exact prefix matching metric.

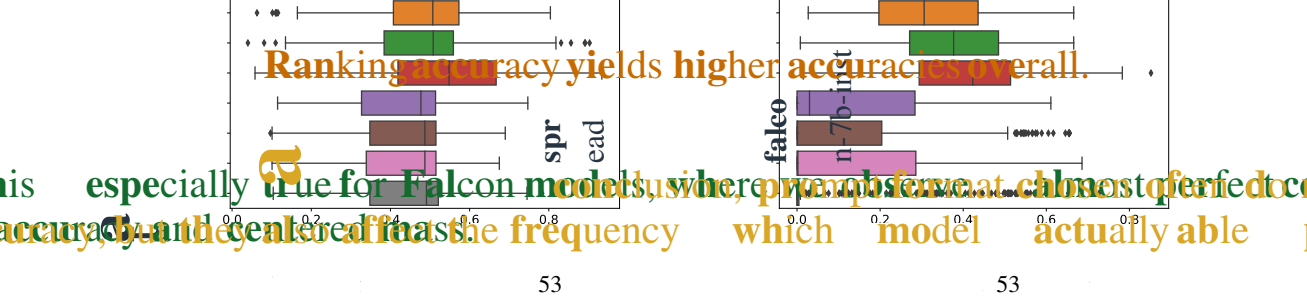
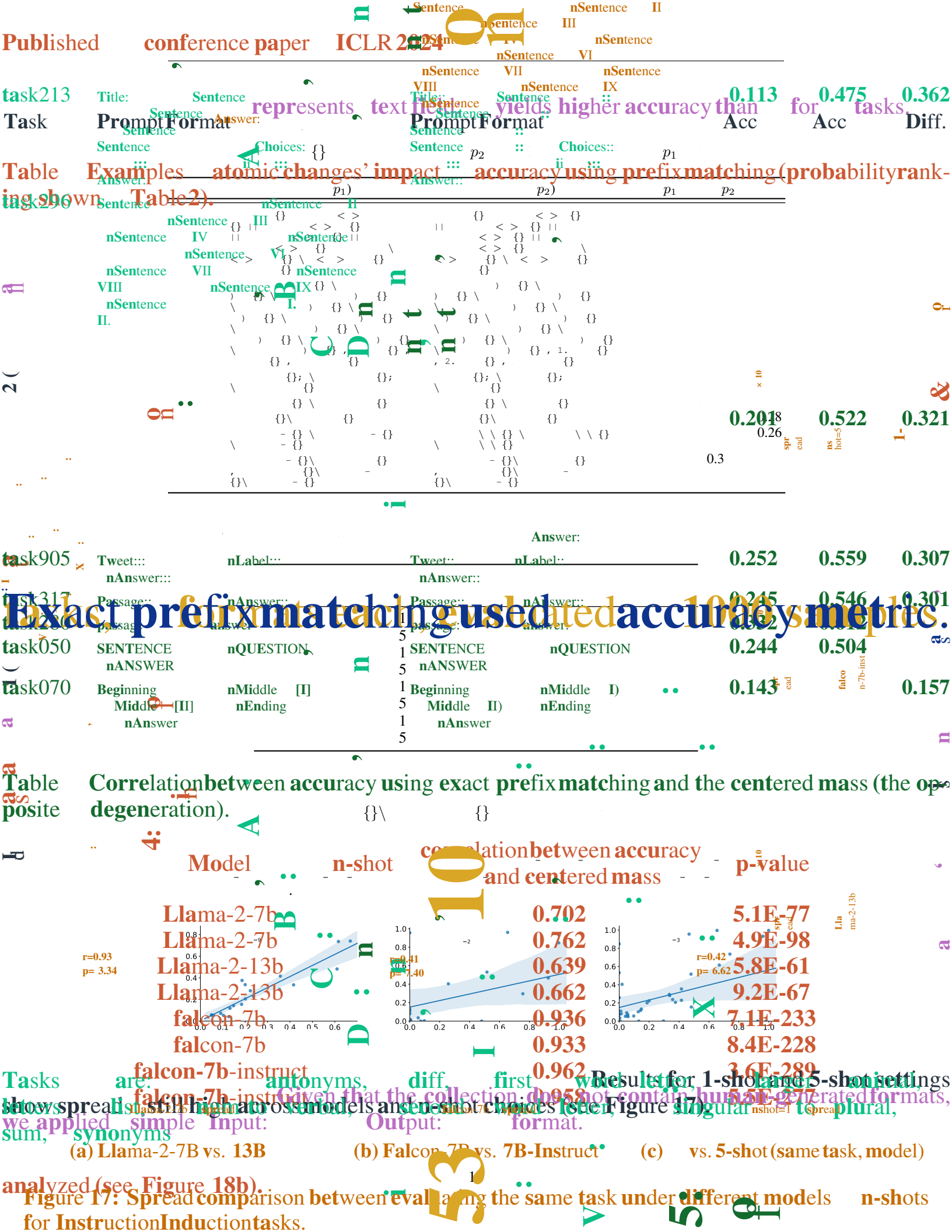


Figure 16: Accuracy metric used, and strongly impact final performance. The boxplots show the distribution of accuracy for different models and shot counts.

This will especially affect tasks for which there are no alternative metrics. We will now focus on tasks that require generating (short) text, and cannot be framed as classification tasks. Examples include identifying Induction tasks also show, a wide range of difficulty, resulting in varied settings.

We selected 10 tasks from Instruction Induction (Honovich et al., 2023) that require generating unique, valid string considered correct response. All experiments thus far focused solely on classification tasks. We selected 10 tasks from Instruction Induction (Honovich et al., 2023) that require generating unique, valid string considered correct response.

Experiments with Instruction Induction tasks. Llama-2-13b



Experiments with continuous metrics. We have shown that spread remains high when considering option ranking, exact prefix matching, accuracy metric.

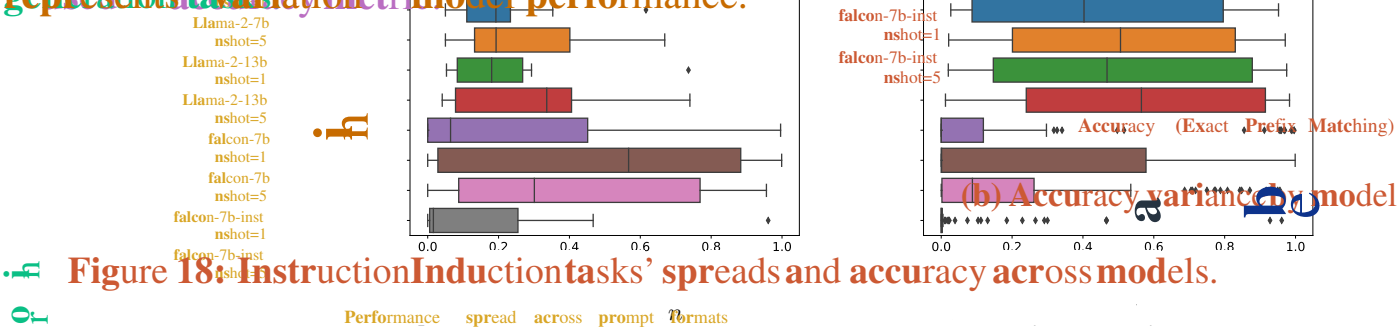


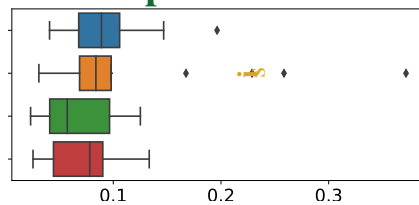
Figure 18: Instruction Induction tasks' spreads and accuracy across models.

(a) Spreads across models and n-shots.

Since LLMs are often used in more open-ended generation contexts, we will now explore the performance variance across prompt formats when considering sentence-length generation tasks (e.g. 0.098 higher, and 0.09 higher. The first n-gram-based metric, and the first n-gram-based metric, and both are metrics where higher values indicate better performance. We leave future work to explore content that each format may induce.

To analyze automatic generations, we use two widely used metrics: ROUGE-L (Lin, 2004), and BERTScore (Ren et al., 2019).

Finally, worth noting, text generation metrics are known to be noisy, and thus not necessarily correspond to true performance loss, the case for accuracy single-valid-output tasks.



We used SuperNatural Instruction tasks: task037, task038,

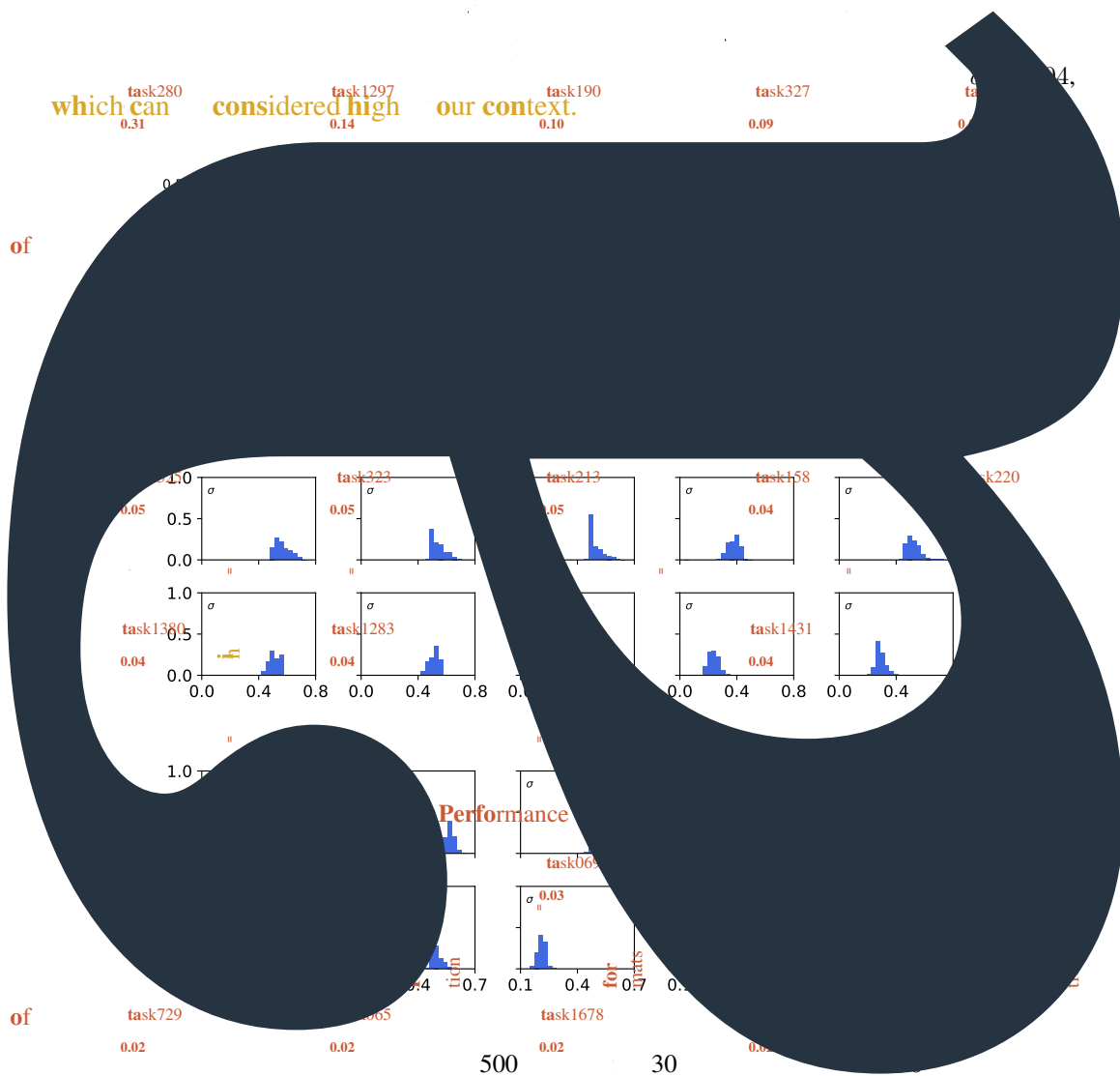
task040, task067, task071, task072, task105, task216, task223, task240, task241, task242, task243, task244, task245, task246, task247, task248, task249, task250, task251, task252, task253, task254, task255, task256, task257, task258, task259, task260, task261, task262, task263, task264, task265, task266, task267, task268, task269, task270, task271, task272, task273, task274, task275, task276, task277, task278, task279, task280, task281, task282, task283, task284, task285, task286, task287, task288, task289, task290, task291, task292, task293, task294, task295, task296, task297, task298, task299, task300, task301, task302, task303, task304, task305, task306, task307, task308, task309, task310, task311, task312, task313, task314, task315, task316, task317, task318, task319, task320, task321, task322, task323, task324, task325, task326, task327, task328, task329, task330, task331, task332, task333, task334, task335, task336, task337, task338, task339, task340, task341, task342, task343, 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task677, task678, task679, task680, task681, task682, task683, task684, task685, task686, task687, task688, task689, task690, task691, task692, task693, task694, task695, task696, task697, task698, task699, task700, task701, task702, task703, task704, task705, task706, task707, task708, task709, task710, task711, task712, task713, task714, task715, task716, task717, task718, task719, task720, task721, task722, task723, task724, task725, task726, task727, task728, task729, task730, task731, task732, task733, task734, task735, task736, task737, task738, task739, task740, task741, task742, task743, task744, task745, task746, task747, task748, task749, task750, task751, task752, task753, task754, task755, task756, task757, task758, task759, task760, task761, task762, task763, task764, task765, task766, task767, task768, task769, task770, task771, task772, task773, task774, task775, task776, task777, task778, task779, task780, task781, task782, task783, task784, task785, task786, task787, task788, task789, task790, task791, task792, task793, task794, task795, task796, task797, task798, task799, task800, task801, task802, task803, task804, task805, task806, task807, task808, task809, task810, task811, task812, task813, task814, task815, task816, task817, task818, task819, task820, task821, task822, task823, task824, task825, task826, task827, task828, task829, task830, task831, task832, task833, task834, task835, task836, task837, task838, task839, task840, task841, task842, task843, task844, task845, task846, task847, task848, task849, task850, task851, task852, task853, task854, task855, task856, task857, task858, task859, task860, task861, task862, task863, task864, task865, task866, task867, task868, task869, task870, task871, task872, task873, task874, task875, task876, task877, task878, task879, task880, task881, task882, task883, task884, task885, task886, task887, task888, task889, task890, task891, task892, task893, task894, task895, task896, task897, task898, 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Figure 19: Spread across n-shots for Llama-2-7b, considering ROUGE-L and BERTScore metrics.

Tasks with high standard deviation reflect highly ill-distributed obtained dissimilar values. When making performance selection, big models that have the moderate standard deviation and the same range, may select wide (uniform) probabilities, obtaining each value the distribution.

We limit generations tokens.

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On factors influencing spread besides prompt formatting.

$r = 0.228$ ($p = 1.4 \times 10^{-7}$)

Figure 20: Accuracy distribution across tasks for LLaMA-2-7B, sorted by standard deviation, decreasing order. We shot option ranking metric

formatting may influence performance variance, but were unable to predict spread. $r = -0.022$ ($p = 0.615$) $r = -0.099$ ($p = 0.024$) $r = 0.559$

We found that the average prompt length task has negligible correlation with performance. When considering each model individually only LLaMA-2-7B with exact prefix matching showed the strongest correlation for option ranking metric, when jointly considering all models and similar to the standard settings had that negligible correlation with spread:

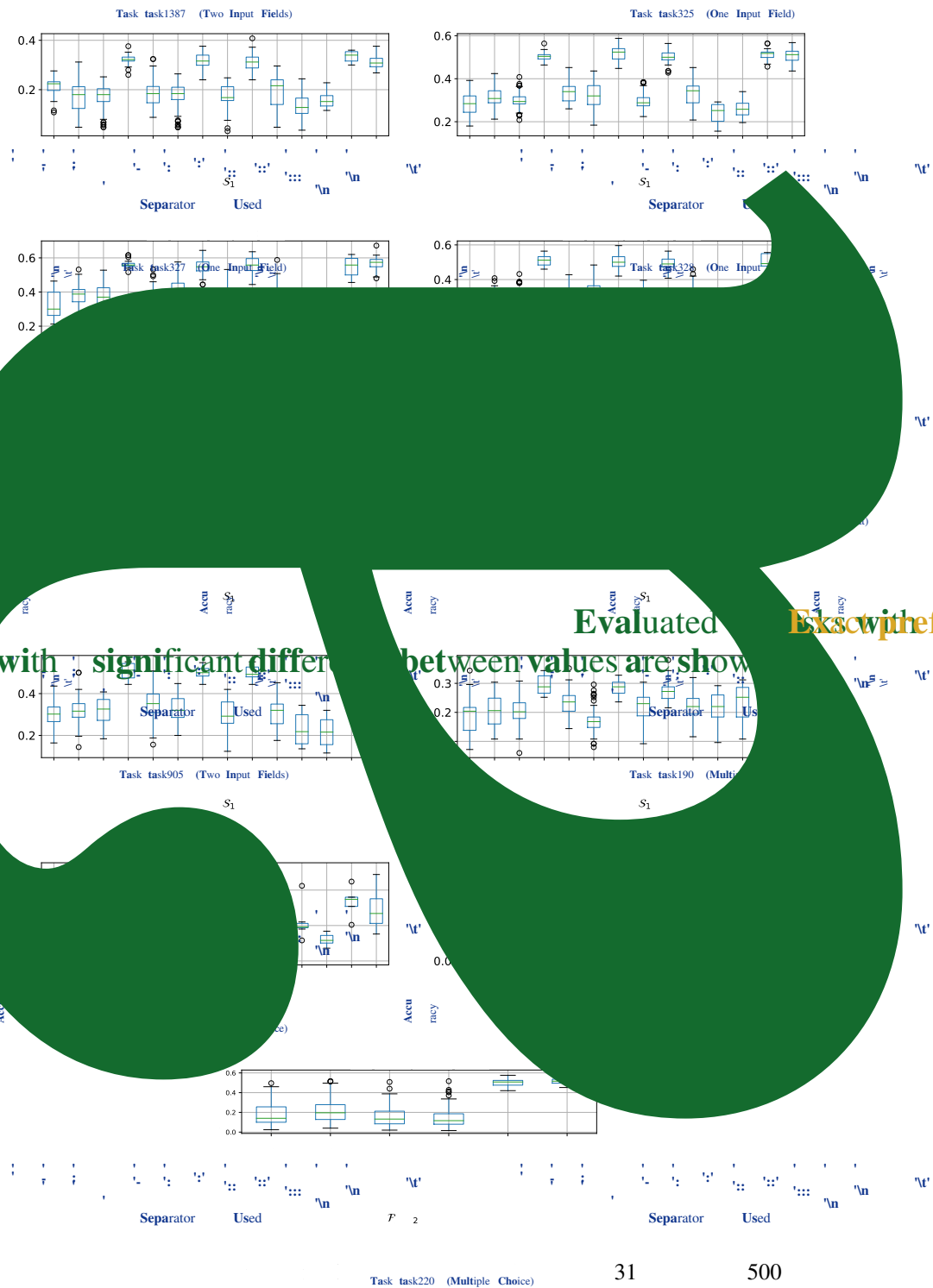
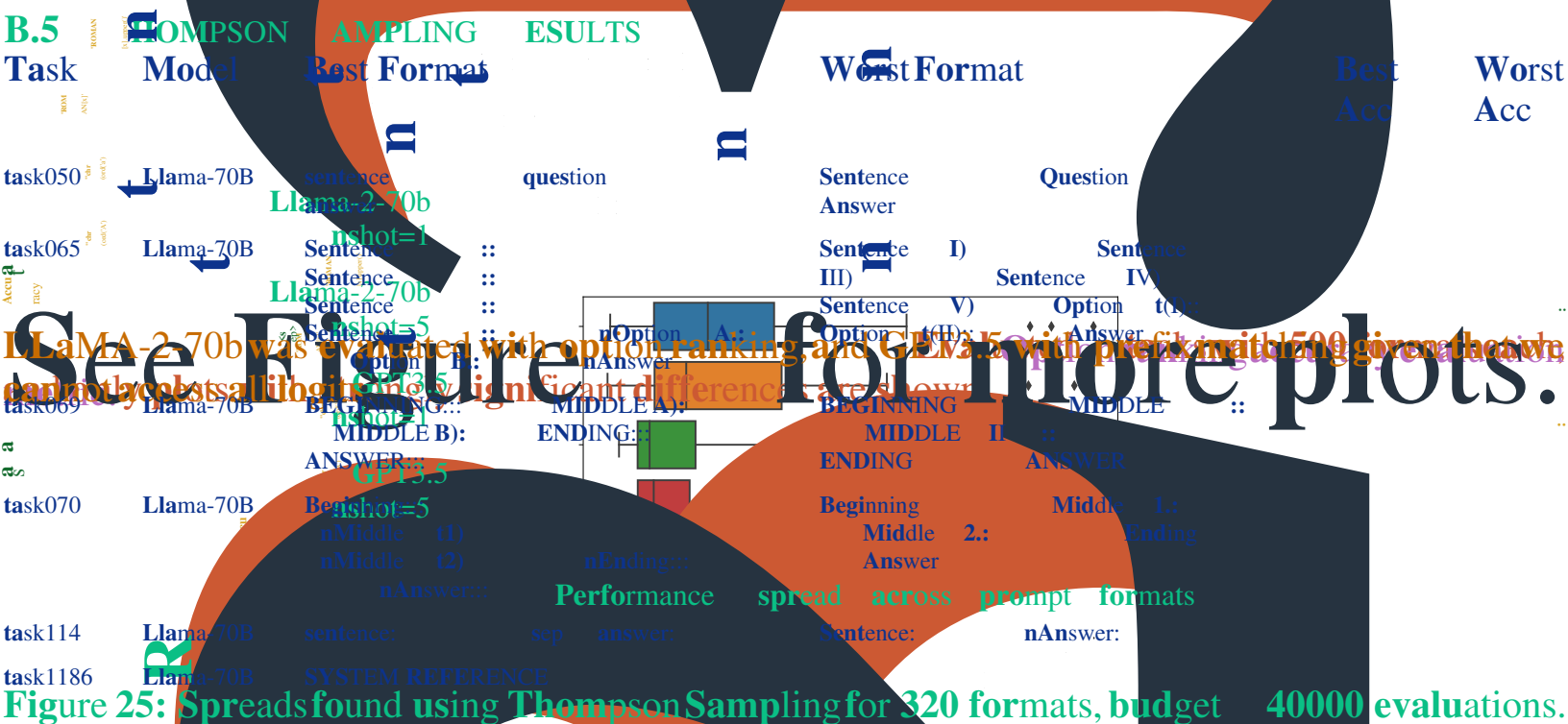
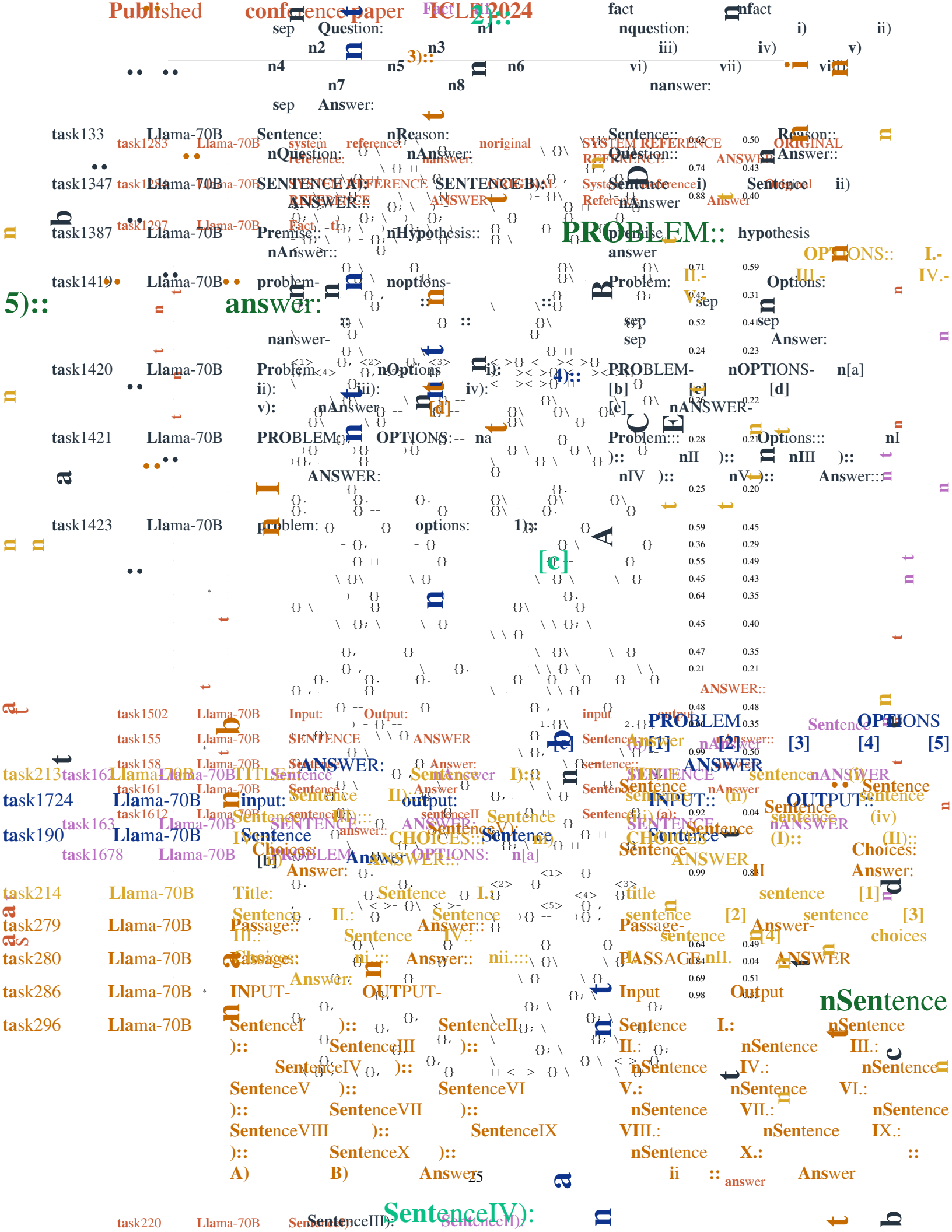
**Figure 22: Variance by feature value for Llama-7B 1-shot.**



Figure 23: Variance by feature value for Llama-7B 1-shot.







Task ID	Model	Input	Output
task220	GPT3.5	Sentence [1]:	Sentence [3]:
task1420	GPT3.5	problem Sentence [4]:	options- (I)
task279	GPT3.5	passage answer Options	Passage n[b]:
task1421	GPT3.5	passage answer Options	Passage n[b]:
task280	GPT3.5	Passage n[A]:	Passage n[b]:
task286	GPT3.5	Input- Output	INPUT:: OUTPUT::
task1423	GPT3.5	problem Sentence	problem Sentence
task1502	GPT3.5	input Sentence	input Sentence
task155	GPT3.5	input Sentence	input Sentence
task158	GPT3.5	input Sentence	input Sentence
task161	GPT3.5	input Sentence	input Sentence
task1612	GPT3.5	input Sentence	input Sentence
task297	GPT3.5	Sentence (A):	Sentence (C):
task162	GPT3.5	Sentence (B):	Sentence (D):
task163	GPT3.5	Sentence (E):	Sentence (F):
task1678	GPT3.5	Sentence (G):	Sentence (H):
task1724	GPT3.5	INPUT:: OUTPUT::	INPUT:: OUTPUT::
task190	GPT3.5	Sentence (I)	Sentence (J)
task316	GPT3.5	passage n[A]:	passage n[B]:
task317	GPT3.5	passage n[A]:	passage n[B]:
task213	GPT3.5	Target:	Target:
task319	GPT3.5	Target:	Target:
task320	GPT3.5	Target::	Target::
task322	GPT3.5	Comment:	Comment:
task214	GPT3.5	Comment:	Comment:

Table 3. Best and worst formats found for task526 and GPT3.5

[illegible]

Tasks marked with asterisk indicate format where the Unicode characters (starting 0x215F), are visualized

using Thompson Sampling 320 formats Llama-70B

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$$B_2^{(2)}(B_1(\quad, \quad, id), B_1(\quad, \quad, id), \quad) \\ B_2^{(2)}(B_1(\quad, \quad, id), B_1(\quad, \quad, id), \quad).$$

IMITATIONS $B_2^{(2)}(B_1(\quad, \quad, id), B_1(\quad, \quad, id), \quad).$

As defined, our grammar all equivalent formats are semantically equivalent human readers. Answer: Yes, Depending we consider the full stop. However, formats may have multiple valid parses, and some may be more prone than others. Contextual restrictions enable Spials to separate undesired combinations, e.g. more spaces and separators, but some values are more unusual, such as the spacing <sep> the separator ::

the format, we may parse the passage as follows: **Passage** 'This sometimes leads equivalent for-
mats to be less usual, yet trivially semantically equivalent results. This last format would have
the following string form: **Contextual relations may also restrict these cases desired the end user.**
input: **Contextual relations may also restrict these cases desired the end user.**

length affects final performance. **Future work may investigate how input**

Less than 100 cases, based on manual inspection of 10 formats and tasks.